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Short Answer Test

1. Computer vision

a) objectives :

* understand images
* Detect and recognize objects
* Support automated decision -making

b) Applications :

* Healthcare
* Autonomous vehicles

2. Features of Computer vision

a) Feature extraction :

Extraction important information like edges , corners , and shapes from an image .

b) object recognition :

Identifies and classifies objects accurately.

3. Levels of image processing

a) low- level image processing :

Improves image quality and using filtering and noise removal .

b) Mid-level vs High level

Image segmentation and scene understanding .

Object recognition and scene understanding .

(4) Digital Image fundamentals

a) pixel :

The smallest unit of a

b) Image resolution :

Higher resolution gives better image quality and more details.

(5) Image Sensing and Acquistion

a) Image sensor :

Converts light into digital signals.

b) Image acquistion devices :

- Digital camera
- webcam

(6) Sampling and Quantization

a) sampling :

Converts a continuous image into discrete pixels.

b) Quantization :

Assigns intensity values to pixels.

(7) Image formation model

a) perspective projection :

Converts a 3D scene into a 2D image.

b) illumination

Provides light needed to capture clear.

8) Image processing vs computer vision

a) Image processing:

Converts a 3D scene enchances or modifies digital images.

b) Two differences:

* Image processing improves images; computer vision understands images.

* Image processing output is information or decisions.

9) Applications

a) Healthcare:

* Detects diseases from X-rays and MRI scans.

b) Autonomous vehicles:

* Detects roads, traffic signs, and pedestrains for safe driving.

10) Digital Image fundamentals

a) pixels:

They are the basic building blocks of digital images.

b) Sampling and resolution

Better sampling and higher resolution improve image analysis accuracy.